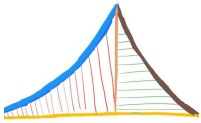
Math for EE



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE-2020

A. Formulae-12TH

FORMULA SHEET

1.1.1 area of a triangle OAB

$$Ar(\triangle OAB) = \frac{1}{2}|\left(OA\right) \times \left(OB\right)| \tag{1.1.1.1}$$

1.1.2

$$sin^{-1}x + cos^{-1}x = \frac{\pi}{2} \tag{1.1.2.1}$$

1.1.3 angle between vectors

$$cos\theta = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right) \tag{1.1.3.1}$$

1.1.4 $A \epsilon R^{n\times n}$

$$|A| = |kI| \implies |A| = k^n|I| \tag{1.1.4.1}$$

1.1.5

$$cos^{-1}(-x) = \pi - cos^{-1}x \tag{1.1.5.1}$$

1.1.6 distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$

$$d = \left|\frac{d_2 - d_1}{\sqrt{a^2 + b^2 + c^2}}\right| \tag{1.1.6.1}$$

1.1.7 king's property

$$\int_a^b f(x)dx = \int_a^b f(a + b - x)dx \tag{1.1.7.1}$$

1.1.8 condition for a function to be continuous at x=a

$$f(a^-) = f(a^+) \tag{1.1.8.1}$$

1.1.9 y is a function of x

$$\frac{dy}{dx} + py = q \tag{1.1.9.1}$$

$$\implies ye^{\int pdx} = \int qe^{\int pdx}dx \tag{1.1.9.2}$$

1.1.10 differentiation of a function $f(x)$ with respect to other function $g(x)$

$$\frac{d(f(x))}{d(g(x))} = \frac{f'(x)}{g'(x)} \tag{1.1.10.1}$$

1.1.11 α, β, γ are the angles made by vector with x,y and z axes respectively

$$cos^2\alpha + cos^2\beta + cos^2\gamma = 1 \tag{1.1.11.1}$$

$$\overrightarrow{a} \cdot \overrightarrow{b} = |a||b|cos\theta \tag{1.1.11.2}$$

1.1.12 characteristic polynomial of a matrix A

$$|A - \lambda I| = 0 \tag{1.1.12.1}$$

1.1.13 integration by parts

$$\int f(x)g(x)dx = f(x)\int g(x)dx - \int f'(x)\left(\int g(x)dx\right)dx \tag{1.1.13.1}$$

1.1.14

$$2sin^{-1}x = sin^{-1}(2x\sqrt{1-x^2}) \tag{1.1.14.1}$$

1.1.15 inverse of a matrix A

$$A^{-1} = \frac{1}{|A|}adj(A) \tag{1.1.15.1}$$

1.1.16 condition for minimum surface area

$$S = \pi r^2 + 2\pi rh \implies \frac{dS}{dr} = 0 \tag{1.1.16.1}$$

$$\implies \frac{d^2S}{dr^2} > 0 \tag{1.1.16.2}$$

1.1.17

$$|\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_1} + \mathbf{c_2}| = |\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_1}| + |\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_2}| \tag{1.1.17.1}$$

1.1.18 Bayes' theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1.1.18.1}$$

B. Problems-12TH

1.2.1 The area of a triangle formed by vertices O, A and B, where $\big(OA\big) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\big(OB\big) = -3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ is

a) $3\sqrt{5}$ sq. units b) $5\sqrt{5}$ sq. units c) $6\sqrt{5}$ sq. units d) 4 sq. units

Solution:
For two given vectors $\big(OA\big)$ and $\big(OB\big)$,

the area of the triangle OAB is given by :
from (1.1.1.1)

$$Ar(\triangle OAB) = \frac{1}{2}\left|\left(\left(OA\right)\times\left(OB\right)\right)\right| \tag{1.2.1.1}$$

so, area of $\triangle OAB$ is

$$\frac{1}{2}\begin{pmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & 2 & 3\\ -3 & -2 & 1\end{pmatrix}$$

$$|4\mathbf{i}-5\mathbf{j}+2\mathbf{k}| = \sqrt{4^2+(-5)^2+2^2}$$

$$Ar(\triangle OAB) = 3\sqrt{5}sq.units$$

.

1.2.2 If $cos(sin^{-1}\frac{2}{\sqrt{5}}+cos^{-1}x)=0$, then x is equal to

a) $\frac{1}{\sqrt{5}}$ b) $-\frac{2}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) 1

Solution:
from (1.1.2.1)

$$cos^{-1}x+sin^{-1}x=\frac{\pi}{2} \tag{1.2.2.1}$$

$cos(\frac{\pi}{2})=0$ so,

$$cos^{-1}\frac{2}{\sqrt{5}}+sin^{-1}x=\frac{\pi}{2}$$

$$x=\frac{2}{\sqrt{5}}$$

.

1.2.3 The interval in which the function f given by $f(x)=x^2e^{-x}$ is strictly increasing is

a) $(-\infty,\infty)$ b) $(-\infty,0)$ c) $(2,\infty)$ d) $(0,2)$

Solution:

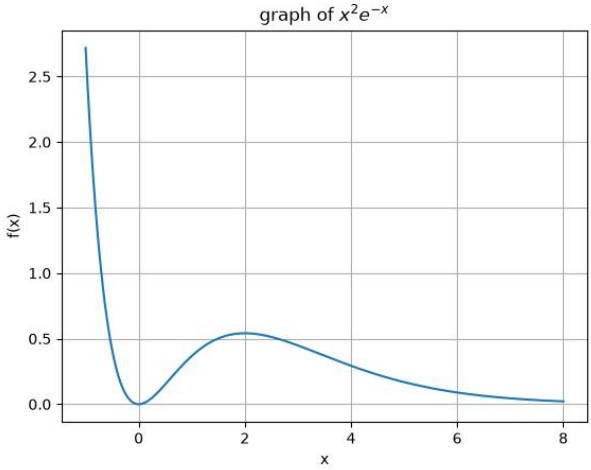
$$f(x)=x^2e^{-x}$$

$$f'(x)=2xe^{-x}-x^2e^{-x}=0 \tag{1.2.3.1}$$

$2x-x^2>0$,for the $f(x)$ to be strictly increasing
so ,

$$x\in(0,2)$$

codes/12/q3.py (1.2.3.2)

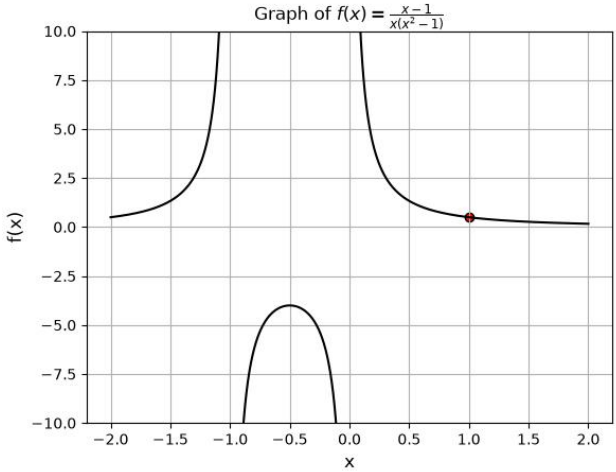


1.2.4 The function $f(x) = \frac{x-1}{x(x^2-1)}$ is discontinuous at

a) exactly one point
b) exactly two points
c) no point
d) exactly three points

Solution:

$f(x)$ domain $\mathbf{R} - \{0, 1, -1\}$, so $f(x)$ is discontinuous at $x = 0, 1, -1$.
3 points of discontinuity



codes/12/q4.py (1.2.4.1)

1.2.5 The function $f : \mathbf{R} \to [-1, 1]$ defined by $f(x) = cos\ x$ is

a) both one-one and onto
b) not one-one, but onto
c) one-one, but not onto
d) neither one-one, nor onto

Solution:
Range of $cosx$ is [-1,1] , so the function $f(x)$ is surjective or onto $cosx$ in $\mathbf{R}$ is not one-one.

codes/12/q5.py (1.2.5.1)

1.2.6 The coordinates of the foot of the perpendicular drawn from the point $(2, -3, 4)$ on the y-axis is

a) $(2, 3, 4)$ b) $(-2, -3, -4)$ c) $(0, -3, 0)$ d) $(2, 0, 4)$

Solution:
foot of perpendicular of the point (x_1, y_1, z_1) on the y-axis is

$$(0, y_1, 0) \implies (0, -3, 0)$$

1.2.7 The relation R in the set 1,2,3 given by $R = \{(1, 2), (2, 1), (1, 1)\}$is

a) symmetric and transitive ,but not reflexive
b) reflexive and symmetric ,but not transitive
c) symmetric ,but neither reflexive nor transitive
d) an equivalence relation

Solution: Relation R is not reflexive ,since $(2, 2), (3, 3) \notin R$
Relation R is not transitive $(2, 1)\epsilon R, (1, 2)\epsilon R$ but $(2, 2) \notin R$
R is symmetric.

1.2.8 The angle between the vectors $\mathbf{i} - \mathbf{j}$ and $\mathbf{j} - \mathbf{k}$ is

a) $-\frac{\pi}{3}$ b) 0 c) $\frac{\pi}{3}$ d) $\frac{2\pi}{3}$

Solution:
angle between two vectors $a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$ and $a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}$ is
from (1.1.3.1)

$$cos\theta = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right) \tag{1.2.8.1}$$

from (1.1.5.1)

$$cos\theta = \left(\frac{-1}{2}\right)$$

so $\theta = \frac{2\pi}{3}$

1.2.9 If A is a non-singular square matrix of order 3 such that $A^2 = 3A$ then value of $|A|$ is

a) -3 b) 3 c) 9 d) 27

Solution:
from (1.1.4.1)

$$|A^2| = |3A|, |A| \neq 0 \tag{1.2.9.1}$$

$$|A| = 27$$

codes/12/q9.py (1.2.9.2)

1.2.10 If $|\mathbf{a}| = 4$ and $-3 \leq \lambda \leq 2$, then $|\lambda\mathbf{a}|$ lies in

a) $[0, 12]$ b) $[2, 3]$ c) $[8, 12]$ d) $[-12, 8]$

Solution:
$-3 \leq \lambda \leq 2$,then $0 \leq |\lambda| \leq 3$
$|\lambda\mathbf{a}|$, $4|\lambda| \in [0, 12]$
so,

$$|\lambda\mathbf{a}| \in [0, 12]$$

1.2.11 If the radius of the circle is increasing at the rate of 0.5 cm/s, then the rate of increase of its circumference is
Solution:
$\frac{dr}{dt} = 0.5cm/s$, circumference $c = 2\pi r$
$\frac{dc}{dt} = 2\pi\frac{dr}{dt}$, $\frac{dc}{dt} = \pi cm/s$

1.2.12 If $\begin{vmatrix} 2x & -9 \\ -2 & x \end{vmatrix} = \begin{vmatrix} -4 & 8 \\ 1 & -2 \end{vmatrix}$ then value of x is
Solution:
$2x^2 - 18 = 0$, $x = \pm 3$

codes/12/q12.py (1.2.12.1)

1.2.13 The corner points of the feasible region of an LPP are $(0, 0)$ $(0, 8)$, $(2, 7)$, $(5, 4)$ and $(6, 0)$ The maximum profit $P = 3x + 2y$ occurs at the point
Solution:
To find the point at which the maximum value of the objective function $P = 3x + 2y$ occurs, we evaluate P at each of the given corner points[cite: 4]:

a) At $(0, 0)$:

$$P = 3(0) + 2(0) = 0$$

b) At $(5,4)$:

$$P = 3(5) + 2(4) = 15 + 8 = 23 \quad \text{(Maximum)}$$

similarly ,we evaluate it at all corners and it is maximum at $(5,4)$
Comparing all the values, the maximum value of P is 23, which occurs at the point $(5,4)$
The maximum profit occurs at the point $(5,4)$

1.2.14 The distance between parallel planes $2x + y - 2z - 6 = 0$ and $4x + 2y - 4z = 0$ is units

Solution:
$4x + 2y - 4z = 12$ and $4x + 2y - 4z = 0$
distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ is
$\mathbf{u} = 4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$
from (1.1.6.1)

$$D = \left| \frac{d_1 - d_2}{||u||} \right| \tag{1.2.14.1}$$

$$D = \left| \frac{12}{\sqrt{4^2 + 2^2 + 4^2}} \right|$$

, D =2 .

or

1.2.15 If P(1,0,-3) is the foot of the perpendicular from origin to the plane , then the cartesian equation of the plane is

Solution: $(OP) = \mathbf{i} - 3\mathbf{k}$ is perpendicular to every point in the plane ,so (OP) can be considered as normal vector of the plane
so equation of the plane

$$1(x-1) + 0(y) - 3(z+3) = 0 \implies x - 3z = 9 \tag{1.2.15.1}$$

1.2.16 Evaluate: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x cos^2 x dx$

Solution:
from (1.1.7.1)

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x cos^2 x dx \tag{1.2.16.1}$$

$$I = \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \frac{\pi}{2} - x \right) cos^2 x dx$$

I = -I ; I=0.

1.2.17 Find the coordinates of the point where the line $\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2}$ cuts the xy-plane.

Solution:
z=0 ,

$$\frac{x-1}{3} = \frac{y+4}{7} = 2$$

$$x = 7, y = 10$$

1.2.18 Find the value of k, so that the function $f(x) = \begin{cases} kx^2 + 5 & if x \leq 1 \\ 2 & if x > 1 \end{cases}$ is continuous at $x = 1$

Solution:

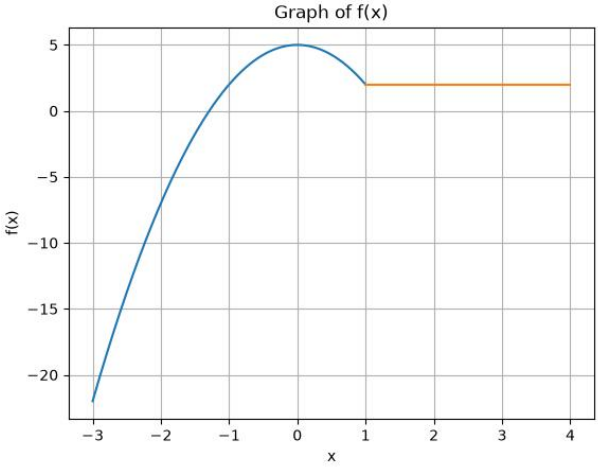
from (1.1.8.1)

$$f(1^-) = f(1^+) \tag{1.2.18.1}$$

$$k + 5 = 2$$

$$k = -3$$

for k=-3,graph of the function would be,



codes/12/q18.py (1.2.18.2)

1.2.19 Find the integrating factor of the differential equation $x\frac{dy}{dx} = 2x^2 + y$

Solution:

from (1.1.9.2)

$$x\frac{dy}{dx} - y = 2x^2 \tag{1.2.19.1}$$

$$\implies \frac{dy}{dx} - \frac{y}{x} = 2x$$

$$ye^{\int -\frac{1}{x}dx} = \int 2xe^{\int -\frac{1}{x}dx}dx$$

integrating factor $e^{\int -\frac{1}{x}dx}$

1.2.20 If $y = f(x^2)$ and $f'(x) = e^{\sqrt{x}}$ then find $\frac{dy}{dx}$

Solution:

$\frac{dy}{dx} = 2xf'(x^2) \implies \frac{dy}{dx} = 2xe^x$

1.2.21 **a.**Find a vector $\mathbf{r}$ equally inclined to the three axes and whose magnitude is $3\sqrt{3}$ units.

Solution:

from (1.1.11.1)

$\alpha = \beta = \gamma = \cos^{-1}(\pm\frac{1}{3})$

$\overrightarrow{r} = 3\sqrt{3}\left(\pm\frac{1}{\sqrt{3}}\hat{i} \pm \frac{1}{\sqrt{3}}\hat{j} \pm \frac{1}{\sqrt{3}}\hat{k}\right)$

or

b.Find the angle between unit vectors $\mathbf{a}$ and $\mathbf{b}$ so that $\sqrt{3}\mathbf{a} - \mathbf{b}$ is also a unit vector.

Solution:

$|\sqrt{3}\mathbf{a} - \mathbf{b}| = \sqrt{3|a|^2 + |b|^2 + 2\sqrt{3}\,(\mathbf{a}\cdot\mathbf{b})}$

$1 = \sqrt{3 + 1 + 2\sqrt{3}\,(\mathbf{a}\cdot\mathbf{b})}$

$\mathbf{a}\cdot\mathbf{b} = -\frac{\sqrt{3}}{2}$

from (1.1.11.2)

$$cos\theta = -\frac{\sqrt{3}}{2} \tag{1.2.21.1}$$

$\implies \theta = \frac{5\pi}{6}$

1.2.22 If $A = \begin{pmatrix} -3 & 2 \\ 1 & -1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ find scalar k so that $A^2 + I = kA$.

Solution:

$|A - \lambda I| = 0 \implies \begin{vmatrix} -3-\lambda & 2 \\ 1 & -1-\lambda \end{vmatrix} = 0$

$\lambda^2 + 4\lambda + 1 = 0 \implies A^2 + I = -4A \implies k = -4$

1.2.23 **a.**If $f(x) = \sqrt{\frac{secx-1}{secx+1}}$, find $f'(\frac{\pi}{3})$

Solution:

$f(x) = \sqrt{\frac{1-cosx}{1+cosx}} \implies f(x) = tan\frac{x}{2}$

$f'(x) = \frac{1}{2}sec^2\frac{x}{2} \implies f'(\frac{\pi}{3}) = \frac{2}{3}$

or

b. Find $f'(x)$ if $f(x) = (tan\ x)^{tan\ x}$

Solution:

from (1.1.13.1)

$$f'(x) = tanx(tanx)^{tanx-1}(sec^2x) + tanx^{tanx}ln(tanx)sec^2x \tag{1.2.23.1}$$

$f'(x) = tanx^{tanx}sec^2x\,(1 + ln(tanx))$

1.2.24 Find $\int \frac{tan^3x}{cos^3x}dx$

Solution:

cosx=t ; sinxdx=-dt

$\int \frac{sin^3x}{cos^6x}dx \implies \int \frac{cos^2x-1}{cos^6x}(-sinxdx) \implies \int \frac{t^2-1}{t^6}dt \implies \int \frac{1}{t^4}dt - \frac{1}{t^6}dt$

$\frac{1}{5cos^5x} - \frac{1}{3cos^3x}$

1.2.25 Show that the plane $x-5y-2z=1$ contains the line $\frac{x-5}{3}=y=2-z$.

Solution:

$\frac{x-5}{3}=y=2-z=k \implies x=3k+5, y=k, z=2-k$

substituting them in the plane equation gives,

$3k+5-5k-2(2-k)-1=0 \implies 0=0$, so the line lies in the plane

1.2.26 A fair dice is thrown two times. Find the probability distribution of the number of sixes. Also determine the mean of the number of sixes.

Solution:

X,Y are the outcomes of each die,

$$P(X,Y)=\begin{cases}\frac{25}{36} & X\neq 6, Y\neq 6\\ \frac{5}{18} & X=6, Y\neq 6 (or) X\neq 6, Y=6\\ \frac{1}{36} & X=6, Y=6\end{cases}$$

mean $\mu=E(x)=\Sigma X, P(X)$

$E(x)=(0*\frac{25}{26})+(1*\frac{5}{18})+(2*\frac{1}{36})$

$$E(x)=\frac{1}{3}$$

1.2.27 Solve the following differential equation

$(1+e^{y/x})dy+e^{y/x}(1-\frac{y}{x})dx=0(x\neq 0)$.

Solution:

Let $\frac{y}{x}=t \implies \frac{dy}{dx}=t+x\frac{dt}{dx}$

$$(1+e^t)(t+x\frac{dt}{dx})+e^t(1-t)=0$$

$$\frac{e^t+t}{e^t+1}dt=-\frac{1}{x}dx$$

$$ln(e^t+t)=ln(\frac{1}{x}) \implies e^t+t=\frac{1}{x}$$

$$e^{\frac{y}{x}}+\frac{y}{x}=\frac{1}{x}$$

1.2.28 Evaluate :

$$\int_0^{\frac{\pi}{2}} sin2xtan^{-1}(sinx)dx$$

Solution:

$I=\int_0^{\frac{\pi}{2}} sin2xtan^{-1}(sinx)dx=\int_0^{\frac{\pi}{2}} sin2xtan^{-1}(cosx)dx$

$sinx=t; cosxdx=dt$

$I=2\int_0^1 ttan^{-1}tdt$

from (1.1.13.1)

$$I=2\left(\left(\frac{t^2}{2}tan^{-1}t\right)_0^1-\int_0^1\frac{t^2}{2(t^2+1)}dt\right) \tag{1.2.28.1}$$

$I = 2\left(\frac{\pi}{8} - \frac{1}{2}\left[t\right]_0^1 + \frac{1}{2}\left[tan^{-1}t\right]_0^1\right)$
$I = \frac{\pi}{2} - 1$

1.2.29 Prove that $tan^{-1}\frac{1}{4} + tan^{-1}\frac{2}{9} = \frac{1}{2}sin^{-1}(\frac{4}{5})$

Solution:
$tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{18}}\right)$
$tan^{-1}(\frac{1}{2}) = sin^{-1}(\frac{1}{\sqrt{5}})$
from (1.1.14.1)

$$2sin^{-1}(\frac{1}{\sqrt{5}}) = sin^{-1}(\frac{4}{5})$$

$\implies sin^{-1}(\frac{1}{\sqrt{5}}) = \frac{1}{2}sin^{-1}(\frac{4}{5})$
Hence proved

1.2.30 Find the equation of the plane passing through the points $(1, 0, -2)$, $(3, -1, 0)$ and perpendicular to the plane $2x - y + z = 8$ Also find the distance of the plane thus obtained from the origin.

Solution:
normal vector of plane 1 i.e, $(2, -1, 1)$ lies on plane 2
dr be the normal vector of plane 2

$$dr_1 = (2, -1, 2); dr_2 = (2, -1, 1)$$

$$dr = dr_1 \times dr_2 \implies (1, 2, 0)$$

$1(x-1) + 2(y) + 0(z+2) = 0 \implies x + 2y = 1$

1.2.31 If $tan^{-1}(\frac{y}{x}) = log\sqrt{x^2+y^2}$, prove that $\frac{dy}{dx} = \frac{x+y}{x-y}$

Solution:
$\frac{y}{x} = t \implies \frac{xdy-y}{x^2} = dt$
$tan^{-1}t = \frac{1}{2}(2logx + log(1+t^2)) \implies \frac{1}{1+t^2}dt = \frac{1}{x} + \frac{tdt}{1+t^2}$

$$\frac{1-t}{1+t^2}dt = \frac{1}{x}$$

$$\frac{x(x-y)}{x^2+y^2}\left(\frac{x\frac{dy}{dx} - y}{x^2}\right) = \frac{1}{x}$$

$$x\frac{dy}{dx} - y = \frac{x^2+y^2}{x-y} \tag{1.2.31.1}$$

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

or

question :
If $y = e^{a\ cos^{-1}x}, -1 < x < 1$, then show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$

Solution:
$\frac{dy}{dx} = -e^{acos^{-1}x}a(\frac{1}{\sqrt{1-x^2}}) \implies \frac{dy}{dx} = \frac{-ay}{\sqrt{1-x^2}} \implies \sqrt{1-x^2}\frac{dy}{dx} = -ay$

$$(1-x^2)(\frac{dy}{dx})^2 = a^2y^2$$

$$2\frac{d^2y}{dx^2}\frac{dy}{dx}(1-x^2) - 2x\left(\frac{dy}{dx}\right)^2 = a^2y^2$$

$$2\frac{d^2y}{dx^2}(\frac{-ay}{\sqrt{1-x^2}})(1-x^2) - 2x\frac{a^2y^2}{1-x^2} = 2a^2y(\frac{-ay}{\sqrt{1-x^2}}) \tag{1.2.31.2}$$

$$(1-x^2)\frac{d^2y}{dx^2} - x\left(\frac{-ay}{\sqrt{1-x^2}}\right) = a^2y$$

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0 \tag{1.2.31.3}$$

1.2.32 Amongst all open (from the top) right circular cylindrical boxes of volume $125\pi\ cm^3$ find the dimensions of the box which has the least surface area.

Solution:

$V = \pi r^2 h \implies h = \frac{125}{r^2}$

The total surface area S

$$S = \pi r^2 + 2\pi rh \tag{1.2.32.1}$$

$$S(r) = \pi r^2 + 2\pi r\left(\frac{125}{r^2}\right) = \pi r^2 + \frac{250\pi}{r} \tag{1.2.32.2}$$

Differentiating with respect to r:

$$\frac{dS}{dr} = 2\pi r - \frac{250\pi}{r^2} \tag{1.2.32.3}$$

from (1.1.16.2)

$$\frac{dS}{dr} = 0$$

:

$$2\pi r - \frac{250\pi}{r^2} = 0 \tag{1.2.32.4}$$

$$2\pi r^3 = 250\pi$$

$$r^3 = 125 \implies r = 5 \text{ cm}$$

Second derivative test:

$$\frac{d^2S}{dr^2} = 2\pi + \frac{500\pi}{r^3} \tag{1.2.32.5}$$

At $r = 5$:

$$\left.\frac{d^2S}{dr^2}\right|_{r=5} = 2\pi + \frac{500\pi}{125} = 2\pi + 4\pi = 6\pi > 0 \tag{1.2.32.6}$$

the surface area is minimum when $r = 5$ cm.
Corresponding height:
$h = \frac{125}{(5)^2} = \frac{125}{25} = 5$ cm
Radius $r = 5$ cm and Height $h = 5$ cm.

1.2.33 Using integration, find the area lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside the parabola $y^2 = 4x$

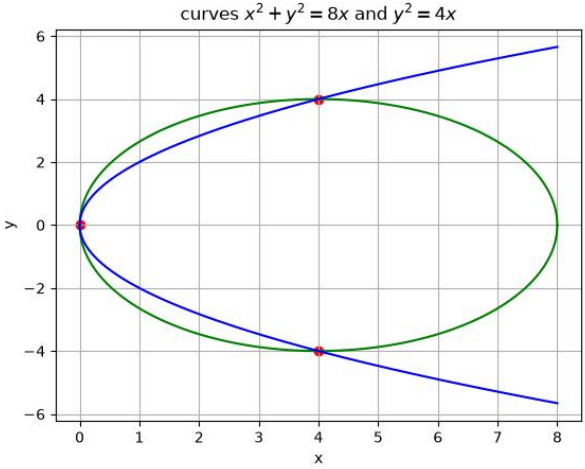
Solution:
Circle equation: $x^2 + y^2 = 8x \implies (x-4)^2 + y^2 = 16$ (Center: $(4, 0)$, Radius: 4)
Parabola equation: $y^2 = 4x$
Substitute $y^2 = 4x$ into the circle equation to find intersection points:
$x^2 + 4x = 8x$
$x^2 - 4x = 0 \implies x(x-4) = 0$
$x = 0$ or $x = 4$
For $x = 0, y = 0$, and for $x = 4, y = \pm 4$.
The required area above the x-axis is split into two regions: 1. From $x = 0$ to $x = 4$, bounded by the parabola $y = 2\sqrt{x}$. 2. From $x = 4$ to $x = 8$, bounded by the circle $y = \sqrt{8x - x^2}$.

$$\text{Area} = \int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{8x - x^2}\,dx$$

$$= 2\left[\frac{x^{3/2}}{3/2}\right]_0^4 + \left[\frac{x-4}{2}\sqrt{8x-x^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_4^8$$

$$= \frac{4}{3}(4^{3/2}) + \left[0 + 8\sin^{-1}(1)\right] - \left[0 + 8\sin^{-1}(0)\right]$$

$$= \frac{32}{3} + 8\left(\frac{\pi}{2}\right) = \left(\frac{32}{3} + 4\pi\right) \text{ sq. units}$$



codes/12/q33.py (1.2.33.1)

or

Question :
Using the method of integration, find the area of the triangle ABC, coordinates of whose vertices are $A(2,0)$, $B(4,5)$ and $C(6,3)$.
Solution:
Equation of line AB passing through $A(2,0)$ and $B(4,5)$:

$$y - 0 = \frac{5-0}{4-2}(x-2)$$

$$y = \frac{5}{2}(x-2) \implies y_1 = \frac{5}{2}(x-2)$$

Equation of line BC passing through $B(4,5)$ and $C(6,3)$:

$$y - 5 = \frac{3-5}{6-4}(x-4)$$

$$y - 5 = \frac{-2}{2}(x-4)$$

$$y - 5 = -(x-4) \implies y_2 = 9 - x$$

Equation of line AC passing through $A(2,0)$ and $C(6,3)$:

$$y - 0 = \frac{3-0}{6-2}(x-2)$$

$$y = \frac{3}{4}(x-2) \implies y_3 = \frac{3}{4}(x-2)$$

$\text{Ar}(\triangle ABC) = \text{Area under } AB\,(x \in [2,4]) + \text{Area under } BC(x \in [4,6]) - \text{Area under } AC$

$$= \int_2^4 \frac{5}{2}(x-2)\,dx + \int_4^6 (9-x)\,dx - \int_2^6 \frac{3}{4}(x-2)\,dx$$

Evaluating the integrals individually:

$$I_1 = \frac{5}{2}\left[\frac{(x-2)^2}{2}\right]_2^4 = \frac{5}{2}\left(\frac{4}{2} - 0\right) = 5$$

$$I_2 = \left[9x - \frac{x^2}{2}\right]_4^6 = (54-18) - (36-8) = 36 - 28 = 8$$

$$I_3 = \frac{3}{4}\left[\frac{(x-2)^2}{2}\right]_2^6 = \frac{3}{4}\left(\frac{16}{2} - 0\right) = 6$$

Combining the results:

$$\begin{aligned}\text{Area}(\triangle ABC) &= I_1 + I_2 - I_3 \\ &= 5 + 8 - 6 \\ &= 7 \text{ sq. units}\end{aligned}$$

area $\implies$ 7 sq. units

1.2.34 If $A = \begin{pmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{pmatrix}$ find A^{-1} and use it to solve the following system of equations:

$5x - y + 4z = 5$

$2x + 3y + 5z = 2$

$5x - 2y + 6z = -1$

Solution:

$$\begin{aligned}|A| &= 5(18-(-10)) - (-1)(12-25) + 4(-4-15) \\ &= 5(28) + 1(-13) + 4(-19) \\ &= 140 - 13 - 76 = 51 \neq 0 \text{ (non-singular)}\end{aligned}$$

Cofactors of A:

$$\begin{aligned} &C_{11} = 28, \quad C_{12} = 13, \quad C_{13} = -19 \\ &C_{21} = -2, \quad C_{22} = 10, \quad C_{23} = 5 \\ &C_{31} = -17, \quad C_{32} = -17, \quad C_{33} = 17\end{aligned}$$

$$\text{Adj}(A) = \begin{pmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{pmatrix} \tag{1.2.34.1}$$

From (1.1.15.1)

$$A^{-1} = \frac{1}{|A|}\text{Adj}(A) = \frac{1}{51}\begin{pmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{pmatrix} \tag{1.2.34.2}$$

The matrix equation is $AX = B$, where $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $B = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$.

$$X = A^{-1}B = \frac{1}{51}\begin{pmatrix} 28(5) - 2(2) - 17(-1) \\ 13(5) + 10(2) - 17(-1) \\ -19(5) + 5(2) + 17(-1) \end{pmatrix}$$
$$= \frac{1}{51}\begin{pmatrix} 140 - 4 + 17 \\ 65 + 20 + 17 \\ -95 + 10 - 17 \end{pmatrix} = \frac{1}{51}\begin{pmatrix} 153 \\ 102 \\ -102 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

codes/12/q34.py (1.2.34.3)

or

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Question : If x, y, z are different and

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

then using properties of determinants show that $1 + xyz = 0$

Solution:

Let $\Delta = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix}$.

Using the property of determinants that allows splitting a column containing a sum of two terms:

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix}$$

Taking out x, y, z common from the $1^{\text{st}}, 2^{\text{nd}}$, and 3rd rows respectively in the second determinant:

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

In the first determinant, interchange columns to match the second determinant:

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = -\begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix} \quad (C_2 \leftrightarrow C_3)$$

$$= (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \quad (C_1 \leftrightarrow C_2)$$

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Now, apply row operations to evaluate the determinant:

$$R_2 \to R_2 - R_1$$

$$R_3 \to R_3 - R_1$$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 0 & y - x & y^2 - x^2 \\ 0 & z - x & z^2 - x^2 \end{vmatrix}$$

Take $(y - x)$ common from R_2 and $(z - x)$ common from R_3:

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y + x \\ 0 & 1 & z + x \end{vmatrix}$$

$$\Delta = (1 + xyz)(y - x)(z - x)[1 \cdot (z + x - (y + x))]$$

$$= (1 + xyz)(y - x)(z - x)(z - y)$$

$$= (1 + xyz)(x - y)(y - z)(z - x)$$

We are given that $\Delta = 0$:

$$(1 + xyz)(x - y)(y - z)(z - x) = 0$$

Since x, y, z are distinct, $(x - y) \neq 0$, $(y - z) \neq 0$, and $(z - x) \neq 0$.
Therefore:

$$1 + xyz = 0$$

1.2.35 A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn randomly one-by-one without replacement and are found to be both kings. Find the probability of the lost card being a king.
Solution:

$$\text{Let } E_1 = \text{Lost card is a king} \implies P(E_1) = \frac{4}{52} = \frac{1}{13}$$

$$\text{Let } E_2 = \text{Lost card is not a king} \implies P(E_2) = \frac{48}{52} = \frac{12}{13}$$

$$\text{Let } A = \text{Two cards drawn from remaining 51 cards are both kings}$$

Using conditional probabilities:

$$P(A|E_1) = \frac{\text{3 kings left}}{\text{51 cards left}} = \frac{3_{C_2}}{51_{C_2}} = \frac{3 \times 2}{51 \times 50} = \frac{6}{2550} = \frac{1}{425}$$

$$P(A|E_2) = \frac{\text{4 kings left}}{\text{51 cards left}} = \frac{4_{C_2}}{51_{C_2}} = \frac{4 \times 3}{51 \times 50} = \frac{12}{2550} = \frac{2}{425}$$

from (1.1.18.1)
Using Bayes' Theorem to find $P(E_1|A)$:

$$\begin{aligned} P(E_1|A) &= \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} \\ &= \frac{\frac{1}{13} \times \frac{1}{425}}{\frac{1}{13} \times \frac{1}{425} + \frac{12}{13} \times \frac{2}{425}} \\ &= \frac{1}{1+24} = \frac{1}{25} \end{aligned}$$

Answer: $\frac{1}{25}$.